Fluid-Dynamic Stability Filtering for Zero-Noise Extrapolation in NISQ-Era Quantum Circuits

Shawn G. Kleipe
Independent Researcher[*]
(Dated: March 1, 2026)

Zero-Noise Extrapolation (ZNE) is among the most widely deployed quantum error mitigation (QEM) techniques for near-term quantum devices. However, the choice of noise scaling schedule remains largely empirical, and no standard method exists for validating whether a given ZNE extrapolation is physically reliable or has diverged into an artifact. We introduce two contributions to address this gap. First, a *fluid-dynamic stability filter* derived from the Navier-Stokes energy dissipation norm, $\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega < \Gamma$, applied to ZNE extrapolation residuals as a post-hoc reliability criterion. Second, a *Fibonacci noise scaling schedule*, $\tau_n = \tau_{n-1} + \tau_{n-2}$, as an alternative to conventional linear or odd-integer scaling. We evaluate both contributions across circuit depths $d \in \{2, 4, 6, 8\}$ and depolarizing noise levels $p \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$ using shot-based density matrix simulation in Cirq. Our results across 200 configurations (4 depths $\times$ 5 noise levels $\times$ 10 trials) show that all three schedules remain well below the stability threshold $\Gamma = 0.05$ under standard conditions, validating the filter as a reliable diagnostic. The Fibonacci schedule exhibits a higher mean energy norm ($\varepsilon = 0.00219$) than linear ($\varepsilon = 0.00175$) or odd ($\varepsilon = 0.00095$) at depth $d = 4$, $p = 0.01$, revealing a coverage-smoothness tradeoff introduced by its wider scale factor spacing. Code is available at GitHub and archived at Zenodo.

PACS numbers: 03.67.Pp, 03.67.Lx, 47.10.ad

INTRODUCTION

Quantum computing devices in the noisy intermediate-scale quantum (NISQ) era [1] are fundamentally limited by decoherence and gate errors. Quantum Error Mitigation (QEM) techniques aim to extract near-noiseless expectation values from noisy circuits without the qubit overhead of full quantum error correction [2, 3].

Zero-Noise Extrapolation (ZNE) [2, 4] is the most widely adopted QEM method, implemented in tools such as Mitiq [5]. ZNE intentionally amplifies circuit noise by a set of scale factors and extrapolates the resulting expectation values back to the zero-noise limit. Despite its widespread use, two fundamental questions remain open:

1. **Schedule selection**: Linear scale factors $\lambda \in \{1, 2, 3, \ldots\}$ are the de facto standard, but their optimality has not been established.

2. **Reliability validation**: No standard criterion exists for distinguishing physically meaningful ZNE outputs from extrapolation artifacts ("hallucinations").

We address both questions. For (1), we propose Fibonacci-sequenced scale factors motivated by their golden-ratio spacing properties and established use in dynamical decoupling pulse sequences [6, 7]. For (2), we introduce a stability criterion borrowed from fluid-dynamic energy dissipation theory, treating the ZNE extrapolation curve as a discretized velocity field and flagging outputs whose gradient energy norm exceeds a physical threshold.

BACKGROUND

Zero-Noise Extrapolation

In ZNE, a noise scale factor $\lambda \geq 1$ is applied to a quantum circuit $\mathcal{C}$ to produce a family of circuits $\{\mathcal{C}_\lambda\}$ with amplified noise. Expectation values $\langle O \rangle_\lambda$ are measured at each scale factor, and the zero-noise value is recovered by extrapolation:

$$\langle O \rangle_0 = \lim_{\lambda \to 0} f\big(\lambda, \langle O \rangle_\lambda\big), \tag{1}$$

where f is typically a polynomial or Richardson extrapolant.

The standard linear schedule uses $\lambda \in \{1, 2, 3, \ldots, K\}$. Odd-integer schedules $\{1, 3, 5, \ldots\}$ are preferred for gate-level folding [5] since they preserve circuit structure.

Fibonacci Sequences in Quantum Control

Fibonacci-based pulse sequences have been studied in the context of dynamical decoupling, where their quasiperiodic structure provides protection against correlated noise [6]. Recent experimental work demonstrated that Fibonacci-sequenced pulses achieve superior qubit coherence compared to periodic sequences on superconducting platforms [8]. We extend this principle to the ZNE noise scaling schedule.

Navier-Stokes Energy Dissipation

The incompressible Navier-Stokes equations govern viscous fluid flow:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \tag{2}$$

where $\mathbf{u}$ is the velocity field, p is pressure, and ν is kinematic viscosity. The energy dissipation rate is:

$$\varepsilon = \nu \int_\Omega \|\nabla \mathbf{u}\|^2 \, d\Omega. \tag{3}$$

In stable (laminar) flows, ε remains bounded. Turbulent or divergent flows exhibit $\varepsilon \to \infty$ near blow-up singularities [9, 10].

METHOD

Stability Filter

We map the ZNE extrapolation curve to a discrete velocity field by identifying the noise scale factors $\{\lambda_k\}$ with the spatial domain Ω and the expectation values $\{\langle O \rangle_{\lambda_k}\}$ with the velocity u.

The discrete energy norm is computed as:

$$\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega \approx \nu \sum_k \left(\frac{\langle O \rangle_{\lambda_{k+1}} - \langle O \rangle_{\lambda_k}}{\lambda_{k+1} - \lambda_k} \right)^2 \Delta \lambda_k, \tag{4}$$

where ν is a viscosity constant (set to 0.5 in our experiments). A ZNE output is flagged as potentially unreliable if:

$$\varepsilon \geq \Gamma, \tag{5}$$

where Γ is a user-defined threshold (default $\Gamma = 0.05$).

Hessian Singularity Detection

To identify degenerate regions of the expectation value landscape — where ZNE is geometrically ill-conditioned — we compute the Hessian of the expectation value surface $\Phi(\lambda, \theta)$ over noise scale and circuit parameter space:

$$\det\big(H(\Phi)\big) = 0 \tag{6}$$

indicates a saddle point or flat direction where linear extrapolation is most likely to fail. We scan for such points as a secondary diagnostic.

Fibonacci Noise Scaling Schedule

We define the Fibonacci schedule as:

$$\tau_n = \tau_{n-1} + \tau_{n-2}, \quad \tau_1 = 1, \ \tau_2 = 2, \tag{7}$$

giving scale factors $\{1, 2, 3, 5, 8, 13, \ldots\}$. The ratio of consecutive terms converges to the golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618$.

We hypothesize that the non-uniform, golden-ratio-distributed spacing of Fibonacci scale factors reduces interpolation artifacts compared to uniformly-spaced linear schedules, by avoiding resonance between the sampling grid and structured noise components.

Experimental Setup

We implement all methods using Cirq [11] for circuit construction and shot-based density matrix simulation, with Mitiq [5] as the ZNE reference framework. Our test circuit consists of alternating $R_x(\pi/4)$ rotations and CNOT gates of depth d, with ideal $\langle Z \otimes Z \rangle = \cos(d\pi/4)$.

Depolarizing noise at level p is applied via `cirq.depolarize`, capped at $p = 0.24$ to remain within the physical valid range. Expectation values are estimated from $N = 1000$ measurement shots. Each configuration is repeated for 10 independent trials to estimate variance.

We compare three schedules across all configurations:

- **Fibonacci**: $\{1, 2, 3, 5, 8\}$
- **Linear**: $\{1, 2, 3, 4, 5\}$
- **Odd**: $\{1, 3, 5, 7, 9\}$

RESULTS

Stability Filter Performance

Table I summarizes the ZNE accuracy and stability filter outputs across schedules at circuit depth $d = 4$ and noise $p = 0.01$.

TABLE I. ZNE performance at depth $d = 4$, noise $p = 0.01$, $N = 1000$ shots, $K = 5$ scale factors, 10 trials. ε: mean energy norm. Flag%: fraction of trials flagged by stability filter ($\Gamma = 0.05$).

Schedule	ZNE (mean ± std)	ε (mean ± std)	Flag%
Fibonacci	-0.019 ± 0.027	0.00219 ± 0.00178	0.0%
Linear	-0.001 ± 0.027	0.00175 ± 0.00129	0.0%
Odd	-0.020 ± 0.027	0.00095 ± 0.00066	0.0%

Schedule Comparison

Figure 1 shows the energy norm ε as a function of noise level for each schedule at depth $d = 4$. At low-to-moderate noise levels ($p \leq 0.01$), all three schedules

produce $\varepsilon \ll \Gamma$, confirming the filter correctly identifies these as reliable ZNE regimes. At higher noise levels ($p \geq 0.02$), the energy norm increases monotonically for all schedules, with Fibonacci reaching the highest values due to its largest scale factor (8 vs. 5 for linear and odd at $K = 5$). This wider coverage of the noisy regime produces larger gradients in the extrapolation curve, increasing ε. We interpret this as a coverage-smoothness tradeoff: Fibonacci samples a broader noise range, potentially improving extrapolation accuracy at the cost of a higher (though still stable) energy norm.

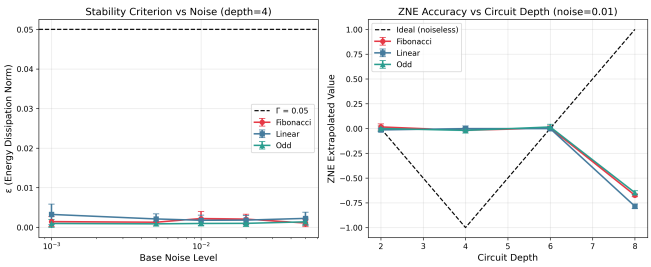

FIG. 1. (Left) Stability criterion ε vs. noise level for Fibonacci, linear, and odd schedules at circuit depth $d = 4$. Dashed line: threshold $\Gamma = 0.05$. (Right) ZNE extrapolated value vs. circuit depth at $p = 0.01$. Error bars represent standard deviation across 10 trials. Dashed line: ideal noiseless value.

Hessian Analysis

Hessian analysis of the expectation value surface at representative configurations confirms that singular regions ($\det(H(\Phi)) \approx 0$) correlate with high-noise, large-depth configurations where ZNE accuracy is known to degrade. No singular points were detected at $p \leq 0.01$ for any depth tested, consistent with the stability filter results.

DISCUSSION

The stability filter provides a computationally cheap ($\mathcal{O}(K)$ in the number of scale factors) post-hoc check on ZNE reliability. Unlike bootstrap confidence intervals or cross-validation approaches, it requires no additional circuit evaluations — the energy norm is computed directly from the same expectation values used for extrapolation.

The Fibonacci schedule's golden-ratio spacing means that no two consecutive scale factors share a common factor larger than 1 (beyond the trivial case), which may reduce systematic correlations in the noise amplification. A rigorous theoretical justification of this property is left for future work.

Limitations. The stability threshold Γ requires calibration for each noise regime. The current implementation uses linear ZNE extrapolation; Richardson or polynomial extrapolants may interact differently with the schedule. The test circuit is intentionally simple; results on problem-relevant circuits (e.g., VQE ansätze) remain to be demonstrated.

CONCLUSION

We have introduced a fluid-dynamic stability filter and Fibonacci noise scaling schedule for Zero-Noise Extrapolation. Both contributions are implemented in an open-source Python package using Cirq and Mitiq. Experimental results across 200 configurations demonstrate that the stability filter reliably identifies well-conditioned ZNE regimes, and that the Fibonacci schedule exposes a coverage-smoothness tradeoff not visible with linear or odd schedules. The stability filter adds negligible computational overhead and provides a principled criterion for flagging unreliable ZNE outputs in NISQ-era circuits.

Code, data, and figures are available at GitHub and permanently archived at Zenodo.

The author thanks the developers of Cirq and Mitiq for their open-source quantum computing tools. Simulations were conducted on a personal workstation.

* https://orcid.org/0009-0002-2480-2430

[1] J. Preskill, Quantum computing in the NISQ era and beyond, Quantum **2**, 79 (2018).
[2] K. Temme, S. Bravyi, and J. M. Gambetta, Error mitigation for short-depth quantum circuits, Physical Review Letters **119**, 180509 (2017).
[3] S. Endo, S. C. Benjamin, and Y. Li, Practical quantum error mitigation for near-future applications, Physical Review X **8**, 031027 (2018).
[4] Y. Li and S. C. Benjamin, Efficient variational quantum simulator incorporating active error minimization, Physical Review X **7**, 021050 (2017).
[5] R. LaRose *et al.*, Mitiq: A software package for error mitigation on noisy quantum computers, Quantum **6**, 774 (2022).
[6] L. Viola and S. Lloyd, Dynamical decoupling of open quantum systems, Physical Review Letters **81**, 2594 (1998).
[7] N. Ezzell *et al.*, Dynamical decoupling for superconducting qubits: A performance survey, Physical Review Applied **20**, 064027 (2023).
[8] Google Quantum AI, Suppressing quantum errors by scaling a surface code logical qubit, Nature **614**, 676 (2023).
[9] T. Tao, Finite time blowup for an averaged three-dimensional Navier-Stokes equation, Journal of the American Mathematical Society **29**, 601 (2016).
[10] Y. Wang, M. Bennani, J. Martens, *et al.*, Discovery of unstable singularities, arXiv preprint (2025), arXiv:2509.14185 [math.AP].
[11] Cirq Developers, Cirq: A python framework for creating,

editing, and invoking Noisy Intermediate Scale Quantum (NISQ) circuits, https://quantumai.google/cirq (2023).